

Astro-1_AA_down BP

nuclear-transcribed mRNA poly(A) tail shortening	piRNA processing	nuclear-transcribed mRNA catabolic process, nonsense-mediated decay	mRNA destabilization	blastocyst formation	dorsal/ventral neural tube patterning	in utero embryonic development
nuclear-transcribed mRNA catabolic process, deadenylation-dependent decay	RNA 3'-end processing	RNA destabilization	nuclear-transcribed mRNA catabolic process	dorsal/ventral neural tube patterning morphogenesis of embryonic epithelium		
positive regulation of mRNA catabolic process	positive regulation of mRNA metabolic process	regulation of mRNA stability	regulation of mRNA catabolic process	blastocyst development	neural tube development	
regulatory ncRNA processing	regulatory ncRNA-mediated gene silencing	regulation of RNA stability	mRNA catabolic process	neural tube formation	embryonic epithelial tube formation	protein localization to ciliary transition zone
cell differentiation in spinal cord	spinal cord motor neuron differentiation	central nervous system neuron development	spinal cord development	neural tube formation	embryonic epithelial tube formation	protein localization to ciliary transition zone
central nervous system neuron axonogenesis	ventral spinal cord development	central nervous system neuron differentiation	telencephalon development	regulation of smoothed signaling pathway	dorsal/ventral pattern formation	

Astro-1_AA_up CC

postsynaptic density membrane		postsynaptic specialization membrane		asymmetric synapse		postsynaptic density		perikaryon					
synaptic membrane		postsynaptic specialization		neuron to neuron synapse		presynaptic endocytic zone membrane		dendritic perikaryon					
						photoreceptor ribbon synapse		neuronal cell body					
postsynaptic membrane		presynaptic membrane		Schaffer collateral – CA1 synapse		presynaptic endocytic zone		neuronal cell body membrane					
				GABA-ergic synapse		ribbon synapse		endocytic vesicle					
cation channel complex		AMPA glutamate receptor complex		ionotropic glutamate receptor complex		voltage-gated calcium channel complex		calcium channel complex		voltage-gated potassium channel complex		endocytic vesicle	
												endocytic vesicle membrane	
monoatomic ion channel complex		transmembrane transporter complex		transporter complex		neurotransmitter receptor complex		potassium channel complex		L-type voltage-gated calcium channel complex		axolemma	
								plasma membrane signaling receptor complex				synaptic cleft	

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Astro-1_AA_up MF

voltage-gated channel activity		voltage-gated monoatomic cation channel activity		voltage-gated monoatomic ion channel activity		metal ion transmembrane transporter activity		phosphatidylinositol bisphosphate binding
gated channel activity		monoatomic ion channel activity		voltage-gated potassium channel activity		potassium channel activity		phosphatidylinositol biphosphate binding
		channel activity		voltage-gated channel activity		calcium ion transmembrane transporter activity		phosphatidylinositol phosphate binding
monoatomic cation channel activity		passive transmembrane transporter activity		potassium ion transmembrane transporter activity		voltage-gated monoatomic ion channel activity involved in regulation of presynaptic membrane potential		phosphatidate phosphatase activity
delayed rectifier potassium channel activity		voltage-gated calcium channel activity		branched-chain amino acid transmembrane transporter activity		neurotransmitter transmembrane transporter activity		phosphatidate phosphatase activity
AP-1 adaptor complex binding		voltage-gated calcium channel activity		amino acid:monoatomic cation symporter activity		amino acid:sodium symporter activity		lipid phosphatase activity
		AP-1 adaptor complex binding		AP-1 adaptor complex binding		AP-1 adaptor complex binding		channel regulator activity

Macro_AA_up CC



Macro_AA_up BP



Macro_AA_up MF

dopamine receptor binding

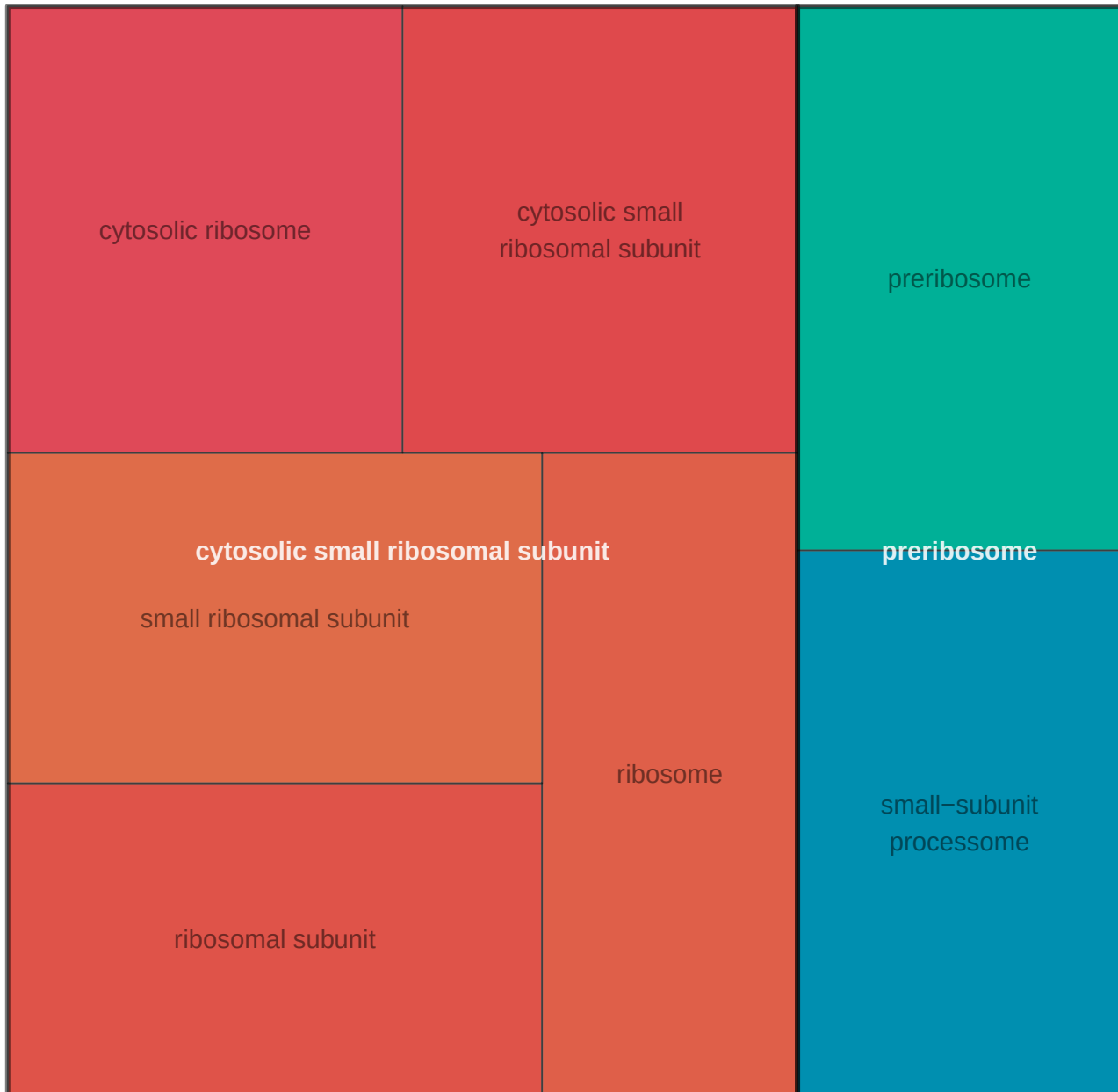
beta-catenin binding

chromatin DNA binding

G-protein
beta/gamma-subunit
complex binding

transcription coregulator binding

Micro-1_AA_down CC



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negative regulation of post-translational protein modification	negative regulation of protein modification by small protein conjugation or removal	negative regulation of protein ubiquitination	neural tube closure	primary neural tube formation	positive regulation of intrinsic apoptotic signaling pathway by p53 class mediator	positive regulation of signal transduction by p53 class mediator
negative regulation of proteolysis involved in protein catabolic process	cytoplasmic translation	regulation of ubiquitin-dependent protein catabolic process	tube closure	embryonic epithelial tube formation	positive regulation of intrinsic apoptotic signaling pathway by p53 class mediator	
negative regulation of ubiquitin-dependent protein catabolic process	negative regulation of protein ubiquitination	regulation of protein modification by small protein conjugation or removal	epithelial tube formation	tube formation	regulation of signal transduction by p53 class mediator	signal transduction by p53 class mediator
negative regulation of proteolysis	regulation of post-translational protein modification	regulation of proteolysis involved in protein catabolic process	neural tube formation			
negative regulation of molecular function	regulation of ubiquitin protein ligase activity	regulation of ubiquitin-protein transferase activity	morphogenesis of embryonic epithelium	mesenchymal cell differentiation	neural crest cell differentiation	
negative regulation of protein ligase activity	negative regulation of NF-kappaB transcription factor activity	negative regulation of transferase activity	neural tube development	mesenchyme development	rRNA processing	rRNA processing
negative regulation of ubiquitin-protein transferase activity	negative regulation of DNA-binding transcription factor activity	negative regulation of catalytic activity	positive regulation of intrinsic apoptotic signaling pathway	regulation of intrinsic apoptotic signaling pathway	ribosomal small subunit biogenesis	protein stabilization
			positive regulation of apoptotic signaling pathway	apoptotic signaling pathway		

Micro-1_AA_down MF

ubiquitin ligase inhibitor activity

ubiquitin-protein transferase regulator activity

ubiquitin-protein
transferase
inhibitor activity

ubiquitin-protein
transferase
regulator activity

NF-kappaB binding

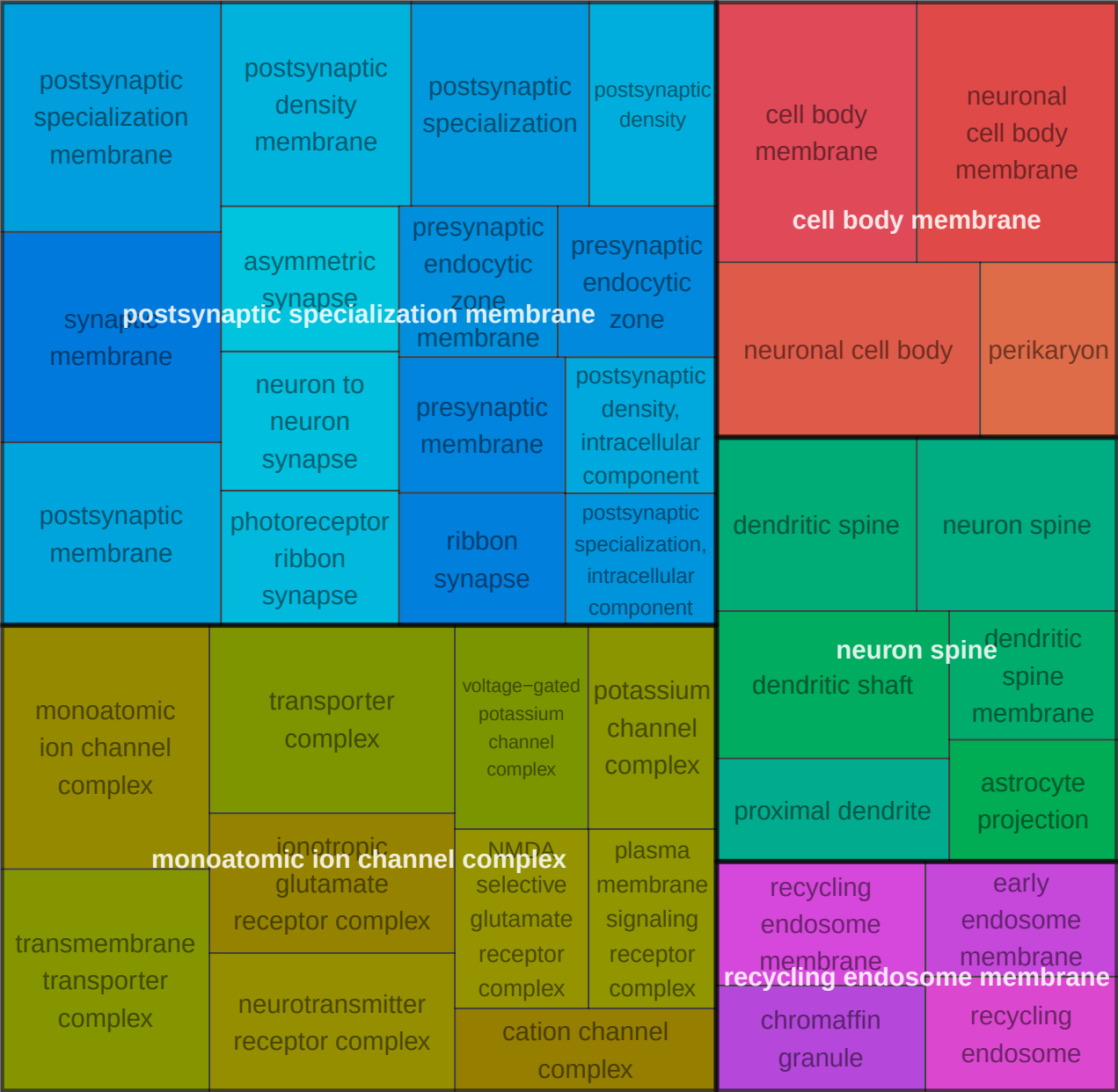
mRNA 5'-UTR binding

mRNA 5'-UTR binding

mRNA 3'-UTR binding

**RNA
methyltransferase
activity**

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Micro-1_AA_up MF

gated channel activity	metal ion transmembrane transporter activity	monoatomic ion channel activity	channel activity		passive transmembrane transporter activity		adenosine receptor binding		beta-2 adrenergic receptor binding		
	glutamate-gated calcium ion channel activity		ligand-gated monoatomic cation channel activity		voltage-gated monoatomic cation channel activity		potassium ion transmembrane transporter activity		adenosine receptor binding		
monoatomic cation channel activity	voltage-gated potassium channel activity	gated channel activity	ligand-gated calcium channel activity		extracellular ligand-gated monoatomic ion channel activity	amino acid transmembrane transporter activity	calcium channel activity		peptide hormone receptor binding	hormone receptor binding	
voltage-gated channel activity			ligand-gated channel activity		calcium ion transmembrane transporter activity	organic acid transmembrane transporter activity	organic anion transmembrane transporter activity	carboxylic acid binding		organic acid binding	
voltage-gated monoatomic ion channel activity	ligand-gated monoatomic ion channel activity involved in regulation of presynaptic membrane potential		glutamate binding					glycine binding			
	potassium channel activity		ligand-gated monoatomic ion channel activity		carboxylic acid transmembrane transporter activity			delayed rectifier potassium channel activity	outward rectifier potassium channel activity		
glutamate receptor activity		neurotransmitter receptor activity		transmitter-gated channel activity		AMPA glutamate receptor activity		G protein-coupled receptor activity involved in regulation of postsynaptic membrane potential			
postsynaptic neurotransmitter receptor activity	glutamate-gated receptor activity		neurotransmitter receptor activity involved in regulation of postsynaptic membrane potential		transmitter-gated monoatomic ion channel activity			NMDA glutamate receptor activity		immunoglobulin binding	
								adenylate cyclase inhibiting G protein-coupled glutamate receptor activity		myosin V binding	
										mRNA regulatory element binding repressor activity	
										G-protein beta-subunit binding	

Micro-3_EA_up CC



Micro-3_EA_up BP



Micro-3_EA_up MF

GABA-gated chloride ion channel activity		extracellular ligand-gated monoatomic ion channel activity		chloride transmembrane transporter activity		monoatomic anion transmembrane transporter activity		adenylate cyclase regulator activity	
ligand-gated monoatomic ion channel activity involved in regulation of presynaptic membrane potential		ligand-gated monoatomic ion channel activity involved in regulation of presynaptic membrane potential		inorganic anion transmembrane transporter activity		ligand-gated monoatomic ion channel activity		cyclase regulator activity	
ligand-gated monoatomic anion channel activity				ligand-gated channel activity				cyclase regulator activity	
G protein-coupled neurotransmitter receptor activity		GABA-A receptor activity		neurotransmitter receptor activity involved in regulation of postsynaptic membrane potential		transmitter-gated channel activity		GABA receptor binding	
G protein-coupled receptor activity				transmitter-gated monoatomic ion channel activity involved in regulation of postsynaptic membrane potential		postsynaptic neurotransmitter receptor activity		G-protein alpha-subunit binding	
				transmitter-gated monoatomic ion channel activity		neurotransmitter receptor activity			

Oligo-1_AA_up CC



Oligo-1_AA_up MF

P-type calcium
transporter activity

P-type ion
transporter
activity

calmodulin binding

P-type calcium transporter activity

P-type transmembrane
transporter activity

ATPase-coupled
monoatomic cation
transmembrane
transporter activity

glutamate receptor binding

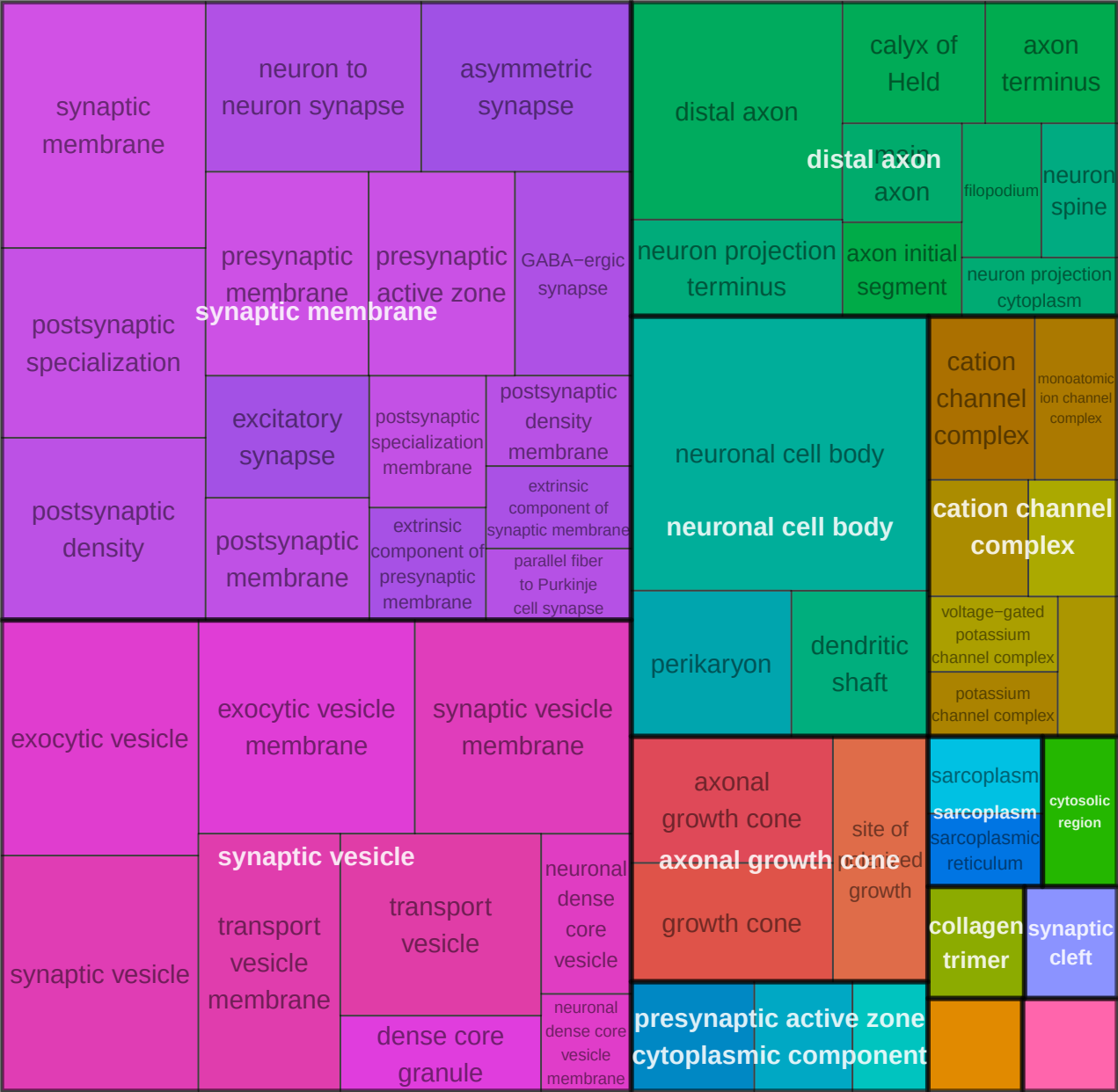
calcium/calmodulin-dependent
protein kinase activity

syntaxin-1 binding

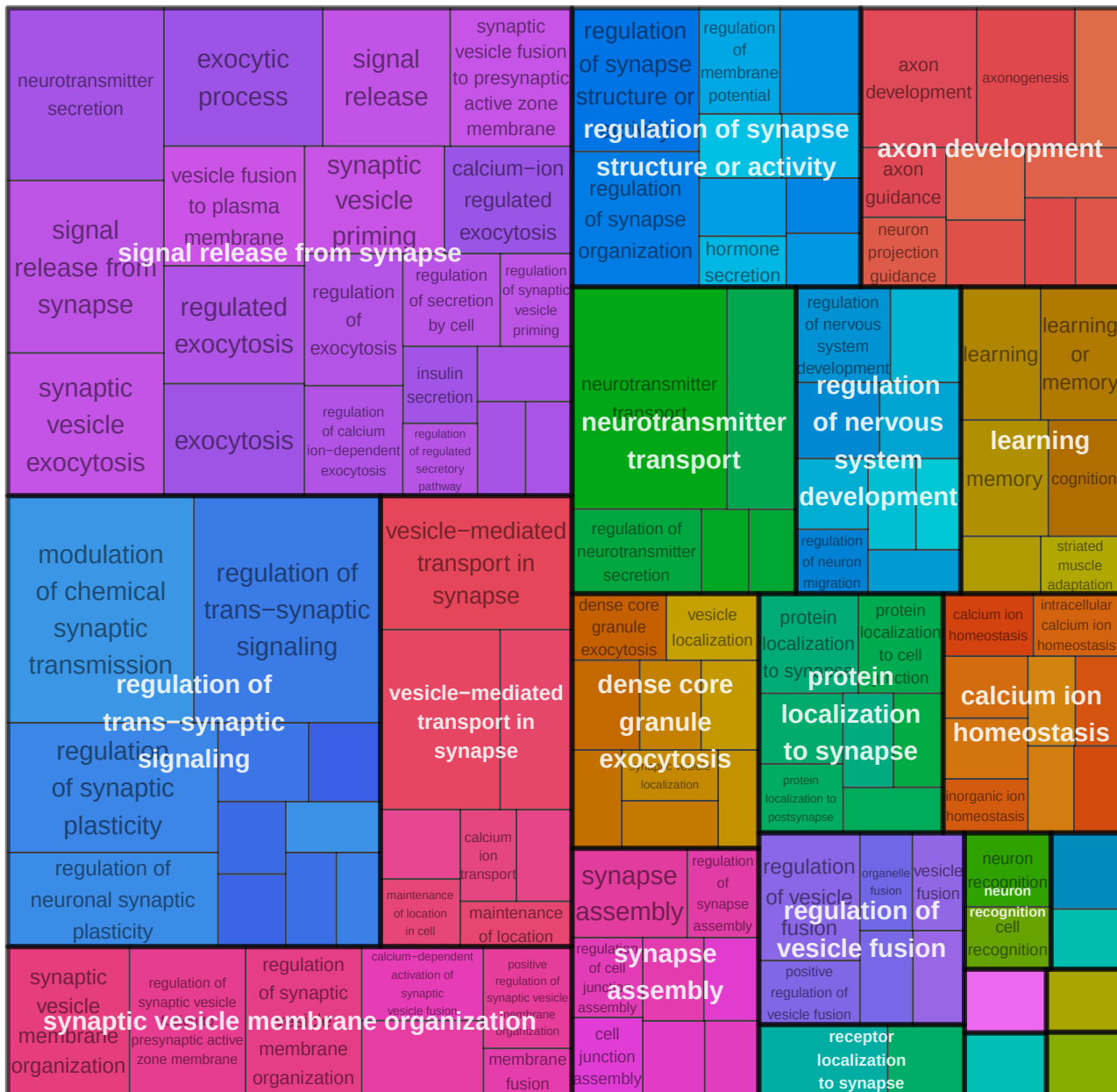
syntaxin-1 binding

diacylglycerol binding

Oligo-3_AA_up CC



Oligo-3_AA_up BP



Oligo-3_AA_up MF



Oligo-3_EA_up CC



Oligo-3_EA_up BP



Oligo-3_EA_up MF

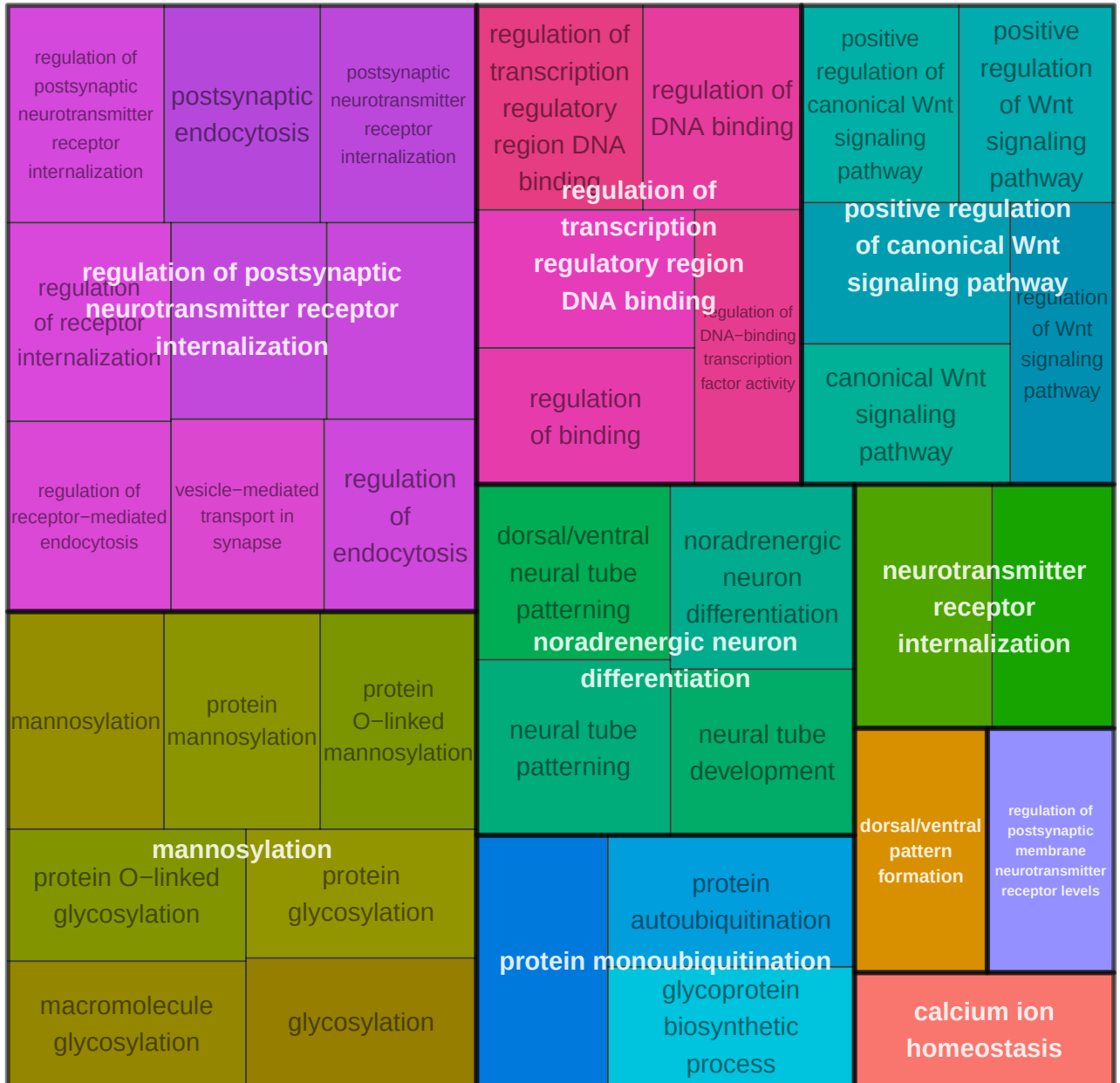
calcium/calmodulin-dependent
protein kinase activity

glycosaminoglycan binding

extracellular matrix structural constituent

growth factor binding

OPC-4_AA_down BP



OPC-4_AA_down MF

dolichyl-phosphate-mannose-protein
mannosyltransferase activity

**dolichyl-phosphate-mannose-protein
mannosyltransferase activity**

mannosyltransferase activity

beta-catenin binding

Vasc-Endo_EA_up CC



Vasc-VLMC_AA_down CC

filopodium tip

filopodium tip

filopodium

actin-based
cell projection

**nuclear
membrane**

perinuclear endoplasmic reticulum

Vasc-VLMC_AA_down BP

